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1 MP-experimental injection (ending the MLIP-energy lottery)

The gen/csp stages used to feed downstream physics (catalysis, battery) whatever the generators produced and the ML hull kept — for known materials that is a lottery: the experimentally relevant polymorph can lose to a generated artifact by MLIP energy error, or never appear at all (olivine vs maricite NaFePO₄ is the canonical case; the paper’s own taxonomy flags it).

Three coordinated mechanisms close every gap between “MP knows the structure” and “the pipeline actually uses it”. All are on by default and degrade to a logged no-op when MP is unreachable.

1.1 1. Experimental GNoME seeds (codes/gnome/templates.py)

build_template_pool gains pool #0: theoretical=False (ICSD-matched) MP entries for the target itself — the exact formula in csp mode, the system and all subsystems in gen mode — with **no ehull cutoff** (the relevant polymorph is often metastable) and a cap (exp_cap, default 10). Highest priority, so the pool dedup keeps these over analog seeds. Knobs: exp_seeds / exp_cap in the GNoME_generate / GNoME_CSP input.yaml blocks.

This makes SAPS build on the right frameworks — but SAPS *transforms* seeds (orbit mapping, substitution scoring, per-template caps), so seeding alone guarantees nothing verbatim. Hence:

1.2 2. Verbatim injection into the relax bundles (csp.py / gen.py)

get_mp_experimental_structures (codes/utis.py) pulls theoretical=False entries sorted by ehull, cap 10 (mp_experimental.cap), and **falls back to the lowest-ehull theoretical entries when a composition has zero experimental hits** — disordered experimental phases (cubic LLZO, LTO, P2-Na layers) are represented in MP by ordered models sometimes flagged theoretical, so experimental-only would lose exactly the solid-state-ionics staples.

The pulled structures ride the existing bundled MLIP relax (bundle order: generated | injected | references, split back by input index via exp_include.split_output_slices — robust to non-converged drops) and are stored with DB source “mp_experimental”. They are relaxed **on-method**, so their energies are directly comparable to everything else on the hull; in gen mode they double as real hull competitors for every chemical subsystem.

1.3 3. Force-include into stable_struct (phase_diagram.py)

In-window injected structures compete on merit like any entry. The force step rescues the rest: stored mp_experimental rows are appended to the manifest even when the ML hull window (EHULL_ML = 0.05 eV/atom) would drop them —

- **deduplicated** against the ML selection *and each other* with exp_include.dedup_forced, so the manifest never carries the same host twice. The dedup matcher is deliberately TIGHTER (pymatgen defaults) than the loose selection matcher: a false merge silently re-loses the experimental polymorph (verified: the loose tolerances merge rocksalt with zincblende; both keep maricite/olivine apart), while a missed merge only costs one redundant host;
- **prepended**, because capped consumers (battery max_hosts) must see the trusted hosts first;

- with their ML `e_above_hull` computed and reported — an “OUTSIDE the ML window, MLIP disagrees with experiment” line in the report is the visible symptom of a model problem, instead of a silently wrong host;
- recorded in the manifest as `stable_struct["forced_experimental"]`.

1.4 input.yaml

```
mp_experimental:      # optional block, defaults shown
  inject: true
  cap: 10
  force_include: true
GNoME_generate:      # (and GNoME_CSP)
  exp_seeds: true
  exp_cap: 10
```

1.5 Guarantees and non-guarantees

With all three mechanisms, every experimentally-known MP polymorph of a submitted composition (cap 10) is in the host set downstream stages see, relaxed on-method, exactly once. What this does NOT fix: structures absent from MP entirely, disordered phases beyond their ordered MP representatives, and MLIP *energies* being wrong — the force-include makes the disagreement visible and survivable, not correct.

Tests: `tests/test_exp_include.py` (pure helpers). Verified live against MP: NaFePO₄ returns maricite + olivine (both experimental, both kept as distinct hosts), LLZO returns the ordered tetragonal garnet.